Chapter 11

# MACRO MODELS AND MULTIPLIERS: LEONTIEF, STONE, KEYNES, AND CGE MODELS

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## 1. INTRODUCTION

A major theme in Erik Thorbecke's career has been extensive work on social accounting matrices (SAMs), both from the perspective of data generation and as providing the underpinnings for a wide range of empirical models. His work is in a long tradition of work with SAMS, including seminal contributions by Richard Stone and Graham Pyatt (a coauthor of Thorbecke's), and a great deal of work by the rest of us. The SAM can be seen both as a data base and as a logical framework for economywide economic models, and Thorbecke has contributed to the development of both.

A SAM can be seen as an extension of Leontief's input-output accounts, filling in the links in the circular flow from factor payments to household income and back to demand for products. The SAM delineates flows across product and factor markets, and provides the statistical underpinnings for multi-sector, multi-factor, computable general equilibrium (CGE) models, much as the national accounts provide the data framework for macro-econometric models. Since a SAM includes all economic flows—indeed, provides the organizing framework for the system of national accounts—and is organized around the accounts of all economic "actors" in the economy, the aggregate national accounts and macro models based on them are necessarily embodied in the SAM framework. The converse is also true: any SAM-based model must, of necessity, incorporate macro aggregates and, implicitly or explicitly, incorporate relationships among them. In particular, CGE models, which are always based on a SAM framework, are theoretically grounded in Walrasian general equilibrium theory, but nonetheless incorporate macro aggregates and macro issues.

There is a longstanding theoretical tension between Walrasian models, with their usual assumptions of market-clearing price-adjustment mechanisms in product and factor markets, and macro models where there is unemployment and very different notions of equilibrium. The literature on CGE models is replete with debates about the macro properties of these models, and a number of different schools of thought have emerged concerning how, or indeed whether, one should incorporate macro features into these SAM-based models. No clear consensus has emerged, which is hardly surprising since the debate really concerns the theoretical divide between Walras and Keynes, and the micro foundations of macro models—or the lack thereof.

In this paper, I discuss a range of models in the SAM framework, from macro-multipliers to CGE models, focusing on the difficulties, pitfalls, and potential insights from different attempts to incorporate macro relationships into SAM-based general equilibrium models. I start with simple multiplier models, and argue that these models capture the essence of the problem of capturing macro phenomena in more elaborate models. I review the various schools of thought, relating the different approaches to the general problem, evident in multiplier models, of specifying macro "closure" in general equilibrium models. Finally, I discuss the theoretical strains that must be accommodated as macro phenomena are incorporated into SAM-based, flow-equilibrium, models.

## 2. SAMS AND MULTIPLIER MODELS

Table 11-1 presents a simple macro SAM—"macro" because it includes only macro aggregates and excludes intermediate inputs (the input-output table). The SAM is square, entries represent payments from column accounts to row accounts, and the corresponding row and column sums must balance since they represent the double-entry, receipt-expenditure accounts of the various economic actors. For the macro SAM, the row and column balances represent the various macro balances in the national income and product accounts. Table 11-2 spells them out.

*Table 11-1.* [Macro SAM]

| | Activities | Commodity | Factors | Hshld | Govt | S-I | World |
|---|---|---|---|---|---|---|---|
| Activities | | D | | | | | E |
| Commodity | | | | C | G | I | |
| Factors | X | | | | | | |
| Household | | | Y | | | | |
| Government | $T^X$ | | | $T^H$ | | | |
| S-I | | | | $S^H$ | $S^G$ | | $S^F$ |
| World | | M | | | | | |

Definitions:

| | |
|---|---|
| D: production sold domestically | C: consumption |
| E: exports | G: government demand |
| X: production (GDP at factor cost) | I: investment demand |
| $T^X$: indirect taxes | $S^H$: household savings |
| $T^H$: direct taxes on households | $S^G$: government savings |
| M: imports | $S^F$: foreign savings |
| Y: factor payments to households | S-I: savings-investment account |

*Table 11-2.* [Balances in the Macro SAM]

1. $GDP = X + T^X = D + E$
2. $D + M = C + G + I$
3. $GDP + (M - E) = C + G + I$
4. $Y = X = GDP$ (factor cost)
5. $Y = C + T^H + S^H$
6. $T^X + T^H = G + S^G$
7. $I = S^H + S^G + S^F$
8. $M = E + S^F$

The SAM is a compact way to present the national accounts, and nicely traces out the circular flow from production activities to factor payments to incomes of “institutions”—households, government, savings-investment (“S-I”)—and back to demand for commodities. GDP at market prices (equation 1, Table 11-2) equals the value of production (*X*, or GDP at factor cost) plus indirect taxes. The “commodity” account represents total “absorption”—the total supply of commodities for use in the economy—and its sum (equations 2 and 3 in Table 11-2) provides a measure of aggregate welfare. As an “actor” in the economy, the commodity account can be seen as a department store which buys domestically produced goods and imports (down the column) and then sells them to other actors in commodity markets (across the row). The “rest of the world” is included as a separate actor, providing imports, buying exports, and financing the difference through foreign savings.

The SAM incorporates the three macro balances: government deficit, trade deficit, and savings-investment balance. The macro balances are expressed as flows—the SAM does not include asset accounts—and any macro relationship in this framework will be in flow terms. In particular, the savings-investment (S-I) account should be seen as representing the "loanable funds" market. The account collects savings from various sources (government, private, and foreign) and spends the accumulated savings on capital goods (*I*). The SAM provides no information about who "owns" the capital goods or in which sectors they are installed. Investment demand in the SAM is by sector of origin, not sector of destination, so the SAM cannot provide information about changes in sectoral capital stocks, or their valuation. While much of macro theory is concerned with the operation of asset markets, SAM-based models need only incorporate the implications for flows across goods and factor markets (commodities and factor services). There is a potential division of labor between modelers working with flows and those working with assets that has been exploited in much of the SAM-based modeling literature, but the division is somewhat artificial—flow and asset market equilibria are obviously linked—and there are theoretical tensions between the two approaches.

All models in the SAM framework must "explain" how balance is achieved in the three macro accounts. Given that the SAM is always balanced, determining two of the macro balances necessarily determines the third. The SAM represents a closed system—all economic transactions are included—and models in this framework will incorporate Walras' Law in some form. They need (indeed, only can) explain one less than the total number of accounts in the SAM. The task facing modelers, then, does not seem very difficult. A model that may include many sectors and factors, and is based on a truly elegant body of general equilibrium theory, must incorporate and determine two out of the three macro balances. Yet how to achieve this apparently simple task has engendered a large and contentious literature.

A very simple approach to modeling in the SAM framework is to expand the SAM to include demands for intermediate inputs and many activities, commodities, factors, and perhaps households; and then to assume that each actor whose account is given in the SAM behaves according to fixed (column) coefficients. For example, producers ("activities") demand inputs in fixed input-output coefficients, commodities are made up of domestically produced and imported goods in fixed proportions, factor income is distributed to households in fixed proportions; households consume and save in fixed shares of total income, and so forth across the SAM. One can interpret these column coefficients as fixed expenditure shares—all behavior is essentially Cobb-Douglas—or assume that prices are fixed exogenously, and so the coefficients also represent real input-output coefficients for the accounts where there is a real counterpart to the expenditures (activities, factors, commodities).

The SAM presented as coefficients describes behavior of the different actors, and provides a description of their behavior across markets. Equilibrium is defined as row-column balance in all SAM accounts. Given fixed prices, row-column balance in commodity and factor markets also represents supply-demand balance, and in the other accounts it represents just receipt-expenditure balance. But not all the accounts can be included in the behavioral model—something has to be made exogenous or the model is over determined. Traditionally, in multiplier models, some of the accounts in the SAM are assumed to be determined exogenously. The SAM is partitioned, separating endogenous and exogenous accounts. The result is a linear model in which the endogenous accounts are a function of the values of the exogenous accounts. Which accounts are specified as endogenous and exogenous determines the macro "closure" of the model.

In the "open" Leontief input-output model, only the activity and commodity accounts are endogenous and all final demand accounts, including exports, are specified as exogenous. In SAM-multiplier models, pioneered by Stone, some combination of the government, S-I, or world accounts is assumed exogenous. In the open input-output model, the column for factor accounts is not needed, since the household account is exogenous. In SAM-multiplier models, it is included since the household account is usually left endogenous.[i] In these models, a change in demand (column entries) by any exogenous account yields equilibrium changes in all endogenous accounts, with changes in the commodity accounts that are generally larger than the exogenous shock—hence the name "multiplier" models. "Equilibrium" in a SAM-multiplier model is defined as row-column (or supply-demand) balance in all endogenous accounts. Given the SAM structure, it must also be true that the solution balances the sum of all the exogenous accounts. The imbalances of these exogenous accounts can be computed from the various SAM coefficients, but their separate balance must be achieved through some unspecified mechanism outside the endogenous part of the multiplier model.

SAM-multiplier models are driven by changes in exogenous demand (column entries) and solve for a resulting change in supply and demand that balances all endogenous accounts. There are no constraints on supply in the model. At fixed prices, all activities are assumed to be able to produce as much as needed to meet the changed demand. Factor demands can be computed from the input-output factor coefficients, but output supply is not constrained by any limits on factor availability.

The richness of SAM multipliers comes from their tracing out chains of linkages from changes in demand to changes in production, factor incomes, household's incomes, and final demands. There is a literature on decomposing these multiplier chains—a literature to which Thorbecke has contributed—and these methods have been widely used to analyze growth linkages in developing countries.[ii]

While the motivation for this work was not to analyze macro equilibrium, the SAM-multiplier models look very much like the simple Keynesian model where unemployment is assumed and output is determined by demand. It is easy to show in simple models that the SAM-multiplier model will generate a demand multiplier equal to one over one minus the marginal propensity to consume—the Keynesian demand multiplier. A change in exogenous demand by the investment account will yield a multiplied increase in income necessary to generate the increased savings to "finance" the additional investment.

The SAM-multiplier models handle achieving macro equilibrium through induced changes in incomes and demand. Prices are fixed, supply is unconstrained, and savings-investment flow equilibrium is achieved through changes in injections into the loanable funds market given fixed savings rates. No supply side, no price adjustments, no assets, no treatment of time or dynamics. Next, we add the supply side in a model with production and prices, but that still focuses on flows—the CGE model.

## 3. THE CLASSIC CGE MODEL AND MACRO BALANCES

A CGE model is Walrasian in spirit, incorporating all the flows in the SAM, including production, distribution, and demand; and determining equilibrium wages and prices by simulating the operation of factor and product markets—the classic neoclassical general equilibrium model of production and exchange. The model can only determine relative prices, and some price or price index is chosen to define the numeraire. The absolute price level is indeterminate and must be specified exogenously. In the classic model, the supply and demand equations are all homogeneous of degree zero in prices—double all prices and equilibrium production and demand does not change—so the absolute price level does not matter to the real side. In macro terminology, the model displays strong neutrality of money. Introducing some mechanism to determine the absolute level of prices such as a simple transactions demand for money plus a fixed money supply would determine the absolute price level, but would not affect relative prices or any real magnitudes.

Typically, classic CGE models specify fixed supplies of primary factors of production (e.g., labor and capital) and assume that all markets "clear" in that prices and wages (defined broadly to include rental rates for all factors) adjust to achieve supply-demand equilibrium in all product and factor markets. In macro terms, the model will always generate full employment of all factors and hence the economy is always operating on the production possibility frontier.[iii] Many applications of CGE models focus on introducing various distortions to the price system and calculating the resulting inefficiencies and loss of welfare. Assuming full employment, however, "inefficiency" is always in terms of being at the wrong place on the production possibility frontier, not from ending up at some point inside the frontier.

## 3.1 Imports, Exports, and the Balance of Trade

Extending the classic Walrasian CGE model to incorporate foreign trade was a major part of the work program in the development of CGE models. Various approaches were tried, but there is now a broad consensus on the general outlines of a "trade focused" CGE model. Such a model incorporates imperfect substitutability between domestically produced and imported goods, citing early work on specifying import demand functions by Paul Armington.[iv] The Armington insight was extended to the treatment of exports, and most trade-focused CGE models specify import demand based on sectoral CES (constant elasticity of substitution) "import aggregation" functions and export supply based on sectoral CET (constant elasticity of transformation) "export transformation" functions. There was some debate about whether this treatment was consistent with neoclassical trade theory, but the model is now widely recognized as being an extension of the Salter-Swan model. It is a theoretically consistent generalization of the "standard" trade model with nontraded goods, introducing degrees of substitutability and transformability rather than assuming a rigid dichotomy between tradable and nontradable goods. The theoretical properties of this model have been worked out in detail.[v]

While exports and imports can be accommodated in the classic CGE model, the introduction of a new actor, the "world", does raise a new problem: What do we do about the balance of trade? Trade theory usually ducks the problem by assuming that it is always zero, but applied models must deal with it since the trade balance is rarely zero in actual data. The simplest solution, widely practiced, is to assume that the trade balance is exogenous and the resulting flow of funds is given to (or taken from) some other actors, typically the savings-investment account. This treatment is shown in the SAM in Table 11-1, where the trade balance is identified as "foreign savings" and added to the S-I account.

The introduction of the trade balance raises a macro issue. Treating it as exogenous means that, for some reason outside the theoretical framework of the CGE model, the budget constraint of one actor includes an exogenous transfer. While transfers among actors are easily incorporated into the classic CGE model, the trade balance is special and is the subject of much study in macro theory. Treating it as an exogenous transfer finesses a lot of questions about why foreigners are willing to hold claims against future exports in a model that does not include assets or time. While the treatment in the CGE model is theoretically coherent, the introduction of the trade balance does raise legitimate concerns among macro economists about the limitations of using only a flow-equilibrium specification.

Adding exports, imports, and the trade balance also raises the issue of how the receipt-expenditure account of the new actor, the world, is brought into balance, or equilibrated. As with the Salter-Swan model, trade-focused CGE models include a new equilibrating variable, the real exchange rate, which is the relative price of aggregates of traded and nontraded goods.

There is an implicit functional relationship between the real exchange rate and the trade balance. Increasing foreign savings always yields an appreciation of the real exchange rate—the price of nontraded goods rises relative to the price of traded goods (exports and imports).[vi] Exports fall as producers shift production toward domestic markets and imports rise as consumers shift demand in favor of imports, bringing the trade balance into equilibrium with the new exogenous higher level of foreign savings.

Most, trade-focused CGE models introduce the exchange rate as an explicit variable, with units of domestic currency per unit of foreign currency. However, the "currency" is not money but simply defines the units of domestic and world prices—domestic prices in local currency units and world prices in foreign currency units (e.g. dollars). The model still contains no assets or money, and the exchange rate is not a "financial" variable in any sense. Changes in the exchange rate work only by changing the relative prices of traded to nontraded goods on domestic markets, affecting export supply and import demand.

The trade-focused CGE model is still very much Walrasian in spirit, determining only relative prices that achieve flow equilibria, and the trade balance can be seen as simply the income-expenditure constraint of the new actor (the "world"). However, that interpretation is strained—the trade balance is also a macro phenomenon, reflecting the operation of asset markets that are outside the CGE theoretical structure. The new actor does not appear to be an optimizing entity in any sense, but simply demands and supplies traded goods (usually at fixed world prices) and finances a specified gap.[vii] The trade balance is a "macro" phenomenon unexplained in the CGE model, and the exchange rate is a "macro" variable in the sense that it is an equilibrating variable indirectly affecting domestic producers and consumers to achieve a specified macro balance—the balance of trade. The exchange rate is not a "signal" in some specific market, especially not any kind of financial market, but operates indirectly though its effect on the relative prices of traded and nontraded commodities.

## 3.2 Savings, Investment, and Government

In addition to the trade balance, CGE models applied to actual economies incorporate savings and the demand for investment goods. The introduction of the S-I account, which collects savings and purchases investment goods, is standard. A new flow equilibrium condition is added to the model—the flow of savings must be made to equal the flow demand for investment goods—and some mechanism is introduced to achieve savings-investment balance. Typically, CGE models specify fixed savings rates by households and assume that whatever is saved is then spent on investment goods. The result is a "savings driven" model of aggregate investment demand.[viii]

As with the introduction of the trade balance, this treatment is theoretically coherent within the flow-equilibrium specification of the CGE model, but raises a host of questions. Why should actors in a static model save? Why should their savings rate be fixed as shares of income? Why should they purchase investment goods rather than increase consumption? Given the common specification of utility functions as depending on consumption, any diversion of funds to purchase investment goods must decrease welfare.[ix] Furthermore, the flow-of-funds specification of the savings-investment account finesses a host of questions. Who owns the new capital stock? Do actors care about asset portfolios?

Savings does not depend on the interest rate or profit rate. Is that a reasonable specification? It is easy to introduce a "loanable funds" market into the CGE model, making saving and investment flows a function of some clearing variable which is interpreted as an interest rate.[x] The model is still expressed in flows, and the clearing "interest rate" variable equilibrates savings-investment flows, not asset portfolios. Incorporating asset markets requires the introduction of time and dynamics, and also financial variables such as money, inflation, and interest rates. Do we need explicit consideration of financial markets? These issues are central to macroeconomics, and it is difficult conceptually to keep them out of a model that includes savings and investment.

The government in a classic CGE model collects taxes, makes and receives transfer payments, and purchases goods and services. It is hard to see the government as being a utility maximizing actor, so most CGE models treat government as following specified rules of behavior.[xi] For example, a common specification is that government expenditure is fixed in real terms, including transfers; government revenue is determined by fixed tax rates; and government savings is determined residually as the gap between revenue and expenditure. Again, this rule-based treatment raises a number of macro questions. The model treats the government deficit or surplus as coming from the loanable funds market. Thus any government deficit "crowds out" private investment. Is this a reasonable specification from a macro perspective? In this flow-of-funds specification, there is no consideration of how the government finances its deficit. No money, no money creation, no financial markets, and no interest rates.

The discussion above has described a typical CGE model that achieves macro balances (or macro "closure") in a particular way. The model assumes full employment, with wages and prices adjusting to achieve equilibrium in factor and product markets. The balance of trade is fixed exogenously, which determines foreign savings. The real exchange adjusts to achieve the specified trade balance through its affect on aggregate imports and exports. The government has a simple rule-based specification: fixed real expenditure, fixed tax rates, and government savings determined residually. Households have fixed savings rates, which determine private savings. Finally, given that all the components of savings are determined by various rules and behavioral parameters, aggregate investment is specified as "savings driven" and equal to the sum of private, government, and foreign savings.

To summarize the discussion so far, the introduction of macro balances and the three deficits into the CGE model is feasible, remaining within the classic general equilibrium paradigm, with its focus on relative prices and flow equilibria. However, theoretical strains are apparent and one can quickly sense the unease of even sympathetic macro economists, and hostility from others, when this model is presented. The research question is: As you introduce macro balances into the classic general equilibrium framework, at what point do you have to open up the model to include "macro" phenomena such as assets, asset markets, money, interest rates, inflation, expectations, and dynamics? And, given the long historical tension between micro and macro theory—Walras versus Keynes—how should one proceed in formulating empirical models?

## 4. RECONCILING CGE AND MACRO MODELS

Classic CGE models do not incorporate the sorts of financial and macro variables that would support analysis of macro stabilization. A CGE model, however, provides a good framework for analyzing issues of structural adjustment: the impact of shocks and policies that work through changing prices and market incentives to affect resource allocation and the structures of demand, production, and trade. Short-run stabilization problems involve links between the financial and real sides of the economy: for example, nominal rigidities, non-neutrality of money, financial credit constraints on production, and so forth. The existence of such real-financial links serious strains the Walrasian paradigm underlying the classic CGE modeling framework.

The debate on how best to use empirical models to analyze both stabilization and adjustment issues is active and far from settled. A number of schools of thought have emerged on how best to adapt and use CGE models for these purposes, and how to reconcile or integrate the CGE and macro modeling frameworks.

**The orthodox school**. This school of thought can be summed up in the maxim: "There is only one model, and its prophet is Walras." In this view, the Walrasian CGE model is theoretically elegant and complete, and any attempt to add macro features and financial variables simply corrupts the model. CGE models should only be used to analyze issues of allocative efficiency, relative prices, and the structure of employment, production, and demand in an environment of well-functioning markets. Macro issues should be left to macro economists.[xii]

**The eclectic school**. In this view, one should build integrated models that incorporate the best elements from Walrasian CGE models and a variety of macro and financial models. The multisector CGE model can be used to provide the supply side of a much richer integrated macro-CGE model that includes assets, including money, asset markets, interest rates, inflation, expectations, and any other features drawn from modern macro theory that seem appropriate to the issues being analyzed.[xiii] These models all embed a CGE model in a dynamic macro model that includes financial variables and asset markets. This literature is very active, and many macro-econometric models now include some kind of CGE model to provide the supply side.

**The ecumenical school.** The philosophy of this school is to use separate CGE and macro-financial models and keep them separate, but specify ways through which the models can talk to one another and cooperate. For example, the models can be linked through variables that are endogenous in one but exogenous in the other.[xiv] A macro model might determine the price level and various macro aggregates as endogenous, and these variables are then specified as exogenous variables in a CGE model. The CGE model, in turn, might determine endogenously variables such as the wage rate and various prices which are taken as exogenous in the macro model. The two models can communicate, and perhaps be solved simultaneously. The advantage of this approach is that the two modeling systems are kept separate, with no need to mix paradigms in a single model.

In the past decade, for good reasons, the influence of the orthodox school has declined, while the eclectic school has clearly grown. First, it has been widely recognized that, as argued above, any applied economywide model must incorporate macro balances and notions of macro equilibrium. While it is possible to include such macro flow equilibria in the Walrasian CGE model, the theoretical strains are serious. If one has to extend the general equilibrium model to incorporate macro phenomena, one might as well go all the way, and incorporate elements of modern macro theory as required. Second, advances in modeling software and solution algorithms have made the specification and solution of forward-looking dynamic CGE models incorporating asset markets and inter-temporal optimization feasible. Third, increased data availability and advances in econometric methods have made it feasible to estimate the parameters of such models. The research frontier in this area is advancing rapidly.

Any attempt to incorporate assets and asset markets into CGE models requires extension of the underlying SAM framework to incorporated asset accounts. Earlier work in this tradition used a "financial SAM" of "FSAM" which extends the flow accounts to include the capital/asset accounts of all major economic actors.[xv] Such an extended data framework is often difficult to implement in practice, since it requires reconciliation of national income and product accounts with financial flow accounts.

While there is active research developing integrated macro-CGE models, the ecumenical approach still has much to recommend it. In particular, extending the classic CGE model to include dynamics and macro features requires opening up the model to incorporate the possibility of unemployment and feedbacks from financial variables to the real side of the economy. The problem is that the classic CGE model, with its assumptions of market clearing and focus on relative prices, is an uneasy host for such macro phenomena. One approach to this problem is to use a separate macro model to determine variables such as aggregate employment, investment, government expenditure, and the three macro balances, and then impose these results onto the CGE model structure. While this "top down" approach is consistent with the ecumenical school, figuring out how to adapt a CGE model to incorporate these effects is also a crucial element in constructing any integrated macro-CGE model.

The eclectic school research program thus needs to build on the ecumenical approach, which can provide examples of ways of adapting the classic CGE model to be an adequate host for including macro phenomena. One can usefully start from the notion of macro closure of the CGE model: "What kinds of macro closures can we specify for the CGE model that will make it behave in ways that reflect the outcomes of macro models." Starting with a simple macro problem, we have come full circle: "How can we make a CGE model behave like a Keynesian multiplier model, where demand elements are an important determinant of aggregate employment and supply?"

## 5. MACRO CLOSURE, MULTIPLIERS, AND CGE MODELS

The issue of macro closure in CGE models brings together the different strands of analysis: SAM multiplier models, CGE models, and macro models.[xvi] At one extreme, it is possible to make a CGE model behave like a fixed-price multiplier model by assuming that primary input prices are all fixed and that supplies of primary inputs are unconstrained—a specification that is reasonable for models of regions within a country.[xvii] Multiplier analysis in such models requires specification of how demand shocks feed through the economy. Similarly, to use a CGE model either with or within a macro model requires extension of the core model to incorporate macro features. Specifying the macro closure of the CGE model is an essential part of that process.

There is a large literature on issues of macro closure of CGE models.[xviii] The essential problem that has to be tackled is that the classic CGE model, in which all markets clear, yields a full-employment equilibrium and market-clearing prices, while short-run macro models typically involve wage and price rigidities, partial adjustment mechanisms, and equilibrium without market clearing, including unemployment. The two paradigms embody very different notions of equilibrium.[xix]

There is a literature on "structuralist" CGE models which embody elements of short-run macro models, including "demand driven" Keynesian equilibria with unemployment.[xx] These models do not explicitly incorporate financial variables and asset markets, but manage to work within the flow-equilibrium structure of CGE models. They effectively impose a macro story onto the CGE model structure—the ecumenical modeling philosophy. To see how these models work, we will consider two simple CGE models, a closed-economy model and an open economy model. For each model, we will consider four alternative macro closures that illustrate how the results from different macro models can be grafted onto the CGE model.

## 5.1 Closed Economy Model

Table 11-3 presents the equations of a one-sector supply-side macro model, Table 11-4 lists the variables, and Table 11-5 presents the SAM associated with the model. The model is "short run" in that capital is assumed to be fixed and labor is the only variable input into production. There are eleven endogenous variables and eleven equations. The model satisfies Walras' Law and thus one equation is redundant. Instead of dropping an equation, we add an additional endogenous variable, *WALRAS*, that must have the value zero at the solution. The top part of Table 11-4 lists the variables that are potentially involved in specifying the macro closure of the model. To define a macro closure, two variables from this list must be specified as endogenous, and all the rest are exogenous.

The first two macro closures in Table 11-4, Neoclassical and Johansen, both assume that both product and labor markets clear, with wages adjusting to yield full employment. In these variants, macro issues are only "compositional" in that they determine the composition of final demand (among C, I, and G), but have no effect on aggregate employment and hence GDP. In the Neoclassical Closure, aggregate investment is determined by aggregate savings, which in turn are determined endogenously through the fixed savings rate out of after-tax income and the government deficit. In Johansen Closure, aggregate investment is assumed to be fixed exogenously, and the savings rate is assumed to adjust to generate the required savings.[xxi] Johansen (1974) explicitly argues that macroeconomic fiscal and monetary polices, presumably outside of the CGE framework, will ensure that savings are generated to "balance" investment.

The second pair of macro closures are more interesting in that they involve Keynesian demand multipliers that affect employment and GDP. In the first Keynesian closure, Keynes 1, the wage is chosen as numeraire, which is common in structuralist macro models by Taylor (1990, 2004). Aggregate investment is fixed exogenously, but instead of adjusting the savings rate, the labor supply is assumed to be endogenous. Adjustment in the real wage is the macro equilibrating mechanism, while the aggregate price level is the equilibrating variable. Assume, starting from equilibrium, that real investment is increased. Savings must be increased to finance the increased investment. The only way that savings can increase is through an increase in income, which requires an increase in employment and output, which requires a fall in the real wage (given that firms are on their demand curve for labor), which in turn requires an increase in the aggregate price level. Assuming no government, this model will behave exactly like the simple Keynesian multiplier model. An increase in investment will yield a new equilibrium level of output which equals 1/(1-mpc) times the increase in investment. Note that the Keynesian multiplier will be exactly the same if the price level is chosen as the numeraire and the wage becomes the macro equilibrating variable. The equilibrating mechanism is the same—the real wage adjusts to generate the employment necessary to generate the income necessary to generate savings equal to aggregate investment.

One problem with the Keynesian macro (multiplier) closure is that any increase in employment is associated with a fall in the real wage—firms are assumed to be on their demand curves for labor, and the wage must fall to induce an increase in labor demand. It has been argued in the macro literature that this result is empirically unrealistic. The Keynes 2 closure specifies a different mechanism. In this case, the price level is chosen as numeraire and the real wage is assumed to be fixed. The aggregate labor supply is assumed to be free, so employment is not fixed. In contrast to Keynes 1 closure, however, firms are assumed not to be on their demand curves for labor. A "labor distortion" parameter, gamma ($\gamma$), is introduced which measures the degree to which the wage deviates from the marginal product of labor. Gamma adjusts so that firms are "induced" (given the first order conditions in equation 2) to hire the labor at the fixed wage necessary to generate the income (and output) necessary to generate the savings necessary to finance investment.

The interpretation of gamma depends on the macro story that is used to justify this mechanism. Following Barro and Grossman (1976), one can assume that product and labor markets are out of equilibrium, and that firms are forced off their labor demand curves. Gamma simply measures how far off they are, but is not a "signal" in any sense in the CGE model. Another interpretation, following Malinvaud (1977), is that firms are demand constrained and rationed in the product market.[xxii] In this interpretation, gamma measures the degree of rationing in the product market. In both cases, the equilibrating mechanism is the same—employment is demand determined, the Keynesian multiplier operates, and the real wage is fixed.

*Table 11-3.* [One Sector Macro Model]

| | |
|---|---|
| $X = F(L, \bar{K})$ | (1) |
| $W_L = \gamma \cdot P^X \cdot \dfrac{\partial F(L, \bar{K})}{\partial L}$ | (2) |
| $W_K \cdot \bar{K} = P^X \cdot X - W_L \cdot L$ | (3) |
| $\tilde{Y}^H = W_K \cdot \bar{K} + W_L \cdot L$ | (4) |
| $\tilde{T} = t^H \cdot Y^H$ | (5) |
| $\tilde{S}^G = \tilde{T} - P^X \cdot G$ | (6) |
| $\tilde{S}^H = MPS^H \cdot \left(\tilde{Y}^H - \tilde{T}\right)$ | (7) |
| $P^X \cdot C = \tilde{Y}^H - \tilde{T} - \tilde{S}^H$ | (8) |
| $P^X \cdot I = \tilde{S}^G + \tilde{S}^H + WALRAS$ | (9) |
| $L = L^S$ | (10) |
| $X = C + I + G$ | (11) |

*Table 11-4.* [Macro Closure Rules for One-Sector Model]

| Variable | Description | Full employment | | Unemployment | |
|---|---|---|---|---|---|
| | | **Neoclassical** | **Johansen** | **Keynes 1** | **Keynes 2** |
| Potential macro closure variables | | | | | |
| $P^X$ | Price of output | Numeraire | Numeraire | | Numeraire |
| $W_L$ | Wage of labor, L | | | Numeraire | Fixed |
| $L^S$ | Supply of labor, L | Fixed | Fixed | | |
| $\gamma$ | Wage distortion | Fixed | Fixed | Fixed | |
| I | Investment demand | | Fixed | Fixed | Fixed |
| $MPS^H$ | Savings rate | Fixed | | Fixed | Fixed |
| $\bar{K}$ | Supply of capital, K | Fixed | Fixed | Fixed | Fixed |
| G | Government demand | Fixed | Fixed | Fixed | Fixed |
| $t^H$ | Tax rate on income | Fixed | Fixed | Fixed | Fixed |
| Other endogenous variables | | | | | |
| X | Output | | | | |
| $L$ | Demand for labor, L | | | | |
| C | Consumption demand | | | | |
| $\tilde{Y}^H$ | Household income | | | | |
| $\tilde{S}^G$ | Government saving | | | | |
| $\tilde{S}^H$ | Household saving | | | | |
| $\tilde{T}$ | Tax revenue | | | | |
| WALRAS | Walras' Law variable | | | | |
| $W_K$ | Wage of capital, K | | | | |

*Table 11-5.* [Social Accounting Matrix (SAM) for One-Sector Model]

| | Expenditures | | | | |
|---|---|---|---|---|---|
| Receipts | Activities | Factors | Household | Savings | Government |
| Activities | | | $P^X \cdot C$ | $P^X \cdot I$ | $P^X \cdot G$ |
| Factors | $W_K \cdot \bar{K} + W_L \cdot L$ | | | | |
| Household | | $\tilde{Y}^H$ | | | |
| Savings | | | $\tilde{S}^H$ | | $\tilde{S}^G$ |
| Government | | | $\tilde{T}$ | | |
| Total | $P^X \cdot X$ | $\tilde{Y}^H$ | $\tilde{Y}^H$ | $P^X \cdot I$ | $\tilde{T}$ |

## 5.2 Open Economy Model

Table 11-6 presents the second model, which adds exports, imports, and foreign savings. Table 11-7 lists the variables, and the model SAM is presented in Table 11-8. The model is an extension of the 1-2-3 model of Devarajan, Lewis, and Robinson (1987, 1990).[xxiii] In the 1-2-3 model (1 country, 2 products, and 3 commodities), the economy produces aggregate output, X (equation 1), which is "transformed" (equation 2) into two products: D which is sold domestically and E which is exported. A third commodity, M, is imported but not produced domestically. The commodities D and M are imperfect substitutes in demand (equation 3), while D and E are imperfectly "transformable" in production delivered to domestic and foreign markets (equation 2). This model is a good simplified representation of most trade-focused CGE models. The 1-2-3-2 version adds a production function and two factors, K and L.

The model has 20 equations and 20 endogenous variables (world prices of exports and imports are assumed fixed). A new macro balance is included, the trade balance or foreign savings, and a new equilibrating variable, the exchange rate. As discussed above, in this class of trade-focused CGE model, there is a relationship between the real exchange rate and the trade balance. Any increase in foreign savings is associated with an appreciation (decrease) in the real exchange rate. Given the new macro balance, to define macro closure, three of the potential closure variables must be specified as endogenous.

As with the closed-economy model, there are two full-employment macro closures and two involving unemployment. In the Neoclassical Closure, the trade balance (foreign savings) is assumed fixed, and the real exchange rate adjusts to achieve equilibrium. In this case, with fixed foreign savings, investment is savings driven, and the model will behave very much like the closed-economy model.

The second closure, "Foreign", is more interesting. In this case, investment is fixed and foreign savings is endogenous. Instead of the domestic savings rate adjusting to achieve savings-investment equilibrium as in Johansen Closure, the equilibrating mechanism is through changes in foreign savings. The macro equilibrating variable is the real exchange rate. A change in investment will induce a change in the real exchange rate to generate a change in the trade balance (foreign savings) to balance the change in investment. The model assumes full employment—the wage varies to clear the labor market, given the exogenous labor supply.[xxiv]

*Table 11-6.* [Model with Factor Markets and Macro Closures]

$$X = F_X(L, \bar{K}) \qquad (1)$$

$$X = G_X(D, E) \qquad (2)$$

$$Q = F_Q(D, M) \qquad (3)$$

$$P^X \cdot X = P^D \cdot D + P^E \cdot E \qquad (4)$$

$$P^Q \cdot Q = P^D \cdot D + P^M \cdot M \qquad (5)$$

$$P^M = R \cdot \bar{\pi}^M \qquad (6)$$

$$P^E = R \cdot \bar{\pi}^E \qquad (7)$$

$$W_L = P^X \cdot \frac{\partial X}{\partial L} \qquad (8)$$

$$W_K = P^X \cdot \frac{\partial X}{\partial K} \qquad (9)$$

$$\frac{E}{D} = g_X(P^D, P^E) \qquad (10)$$

$$\frac{M}{D} = f_Q(P^D, P^M) \qquad (11)$$

$$\tilde{Y}^H = W_L \cdot L + W_K \cdot \bar{K} \qquad (12)$$

$$\tilde{T} = t^H \cdot \tilde{Y}^H \qquad (13)$$

$$\tilde{S}^G = \tilde{T} - P^Q \cdot G \qquad (14)$$

$$\tilde{S}^H = MPS^H \cdot (\tilde{Y}^H - \tilde{T}) \qquad (15)$$

$$P^Q \cdot C = \tilde{Y}^H - \tilde{T} - \tilde{S}^H \quad (16)$$

$$P^Q \cdot I = \tilde{S}^H + \tilde{S}^G + R \cdot S^F + WALRAS \quad (17)$$

$$\bar{\pi}^M \cdot M - \bar{\pi}^E \cdot E = S^F \quad (18)$$

$$L = L^S \quad (19)$$

$$Q = C + I + G \quad (20)$$

*Table 11-7* [Macro Closure Rules for 1-2-3-2 Model]

| | | Full employment | | Unemployment | |
|---|---|---|---|---|---|
| **Variable** | **Description** | **Neoclassical** | **Foreign** | **Keynes 1** | **Keynes 3** |
| Potential macro closure variables | | | | | |
| $P_X$ | Price of X | Numeraire | Numeraire | | |
| $W_L$ | Wage of L | | | Numeraire | Numeraire |
| R | Exchange rate | | | | Fixed |
| I | Investment demand | | Fixed | Fixed | Fixed |
| $S^F$ | Trade balance | Fixed | | Fixed | |
| $L^S$ | Supply of labor | Fixed | Fixed | | |
| G | Government demand | Fixed | Fixed | Fixed | Fixed |
| $\bar{K}$ | Capital in X | Fixed | Fixed | Fixed | Fixed |
| $MPS^H$ | Savings rate | Fixed | Fixed | Fixed | Fixed |
| $t^H$ | Tax rate on income | Fixed | Fixed | Fixed | Fixed |

*Table 11-7* [Cont.]

| | | Full Employment | | Unemployment | |
|---|---|---|---|---|---|
| **Variable** | **Description** | **Neoclassical** | **Foreign** | **Keynes 1** | **Keynes 3** |
| Other variables | | | | | |
| X | Aggregate output | | | | |
| $D$ | Domestic good | | | | |
| $E$ | Export good | | | | |
| $M$ | Import good | | | | |
| $Q$ | Composite good | | | | |
| $L$ | Labor in X | | | | |
| $P_Q$ | Price of Q | | | | |
| $P_D$ | Price of D | | | | |
| $P_M$ | Price of M | | | | |
| $P_E$ | Price of E | | | | |
| $\bar{\pi}_M$ | World price of M | Fixed | Fixed | Fixed | Fixed |
| $\bar{\pi}_E$ | World price of E | Fixed | Fixed | Fixed | Fixed |
| $W_K$ | Wage of K | | | | |
| C | Consumption demand | | | | |
| $\tilde{Y}^H$ | Household income | | | | |
| $\tilde{S}^G$ | Government saving | | | | |
| $\tilde{S}^H$ | Household saving | | | | |
| $\tilde{T}$ | Tax revenue | | | | |
| WALRAS | Walras' Law variable | | | | |

*Table 11-8* [SAM for 1-2-3-2 Model]

| | Expenditures | | | | | | |
|---|---|---|---|---|---|---|---|
| Receipts | Activities | Com | Fctrs | Hshld | S-I | Govt | World |
| Activities | | $P^D \cdot D$ | | | | | $P^E \cdot E$ |
| Commodity | | | | $P^Q \cdot C$ | $P^Q \cdot I$ | $P^Q \cdot G$ | |
| Factors | $W_K \cdot \overline{K} + W_L \cdot L$ | | | | | | |
| Household | | | $\tilde{Y}^H$ | | | | |
| Saving (S-I) | | | | $\tilde{S}^H$ | | $\tilde{S}^G$ | $R \cdot S^F$ |
| Government | | | | $\tilde{T}$ | | | |
| World | | $P^M \cdot M$ | | | | | |
| Total | $P^X \cdot X$ | $P^Q \cdot Q$ | $\tilde{Y}^H$ | $\tilde{Y}^H$ | $P^Q \cdot I$ | $\tilde{T}$ | |

The first Keynesian Closure, Keynes 1, operates in the same manner as in the closed-economy model. The trade balance is assumed to be fixed exogenously, with the real exchange rate adjusting to achieve exports and imports consistent with external equilibrium. Changes in the price level change the real wage, generating employment, output, income, and savings consistent with the specified aggregate investment. Since foreign savings are fixed, they play no role in this adjustment.[xxv] The Keynesian multiplier will operate pretty much the same as in the closed-economy model. In this variant, as in the closed economy, any increase in employment will be associated with a fall in the real wage.

The second Keynesian Closure, Keynes 3, adds a new twist. In this closure, which follows the treatment in a structuralist macro-CGE model of Brazil, the wage is chosen as numeraire, the exchange rate is set exogenously, and foreign savings is endogenous.[xxvi] In this model, as the price level adjusts to change the real wage, as in the Keynesian equilibrating mechanism, the real exchange rate also changes. An increase in the price level lowers the real wage, increasing employment, income, and savings. However, the price rise also appreciates the real exchange rate, increasing imports, reducing exports, and thus inducing an increase in foreign savings. Achieving macro equilibrium after an increase in aggregate investment is now achieved through two sources of savings: increased income generating higher savings and increased foreign savings. The result is a much smaller Keynesian income-employment multiplier.[xxvii]

## 6. CONCLUSION

There are a few lessons from the literature on SAM multipliers and macro closure in CGE models.

Any realistic economywide model must include macro aggregates and some mechanism for achieving macro equilibrium. The SAM structure is a powerful framework for delineating the links between micro and macro "actors" and provides the underlying data for any micro-macro modeling, including multiplier models.

There are strong links between structuralist macro models and SAM multiplier models. SAM multiplier models represent one end of a continuum, a fix-price model that is completely driven by demand. At the other end is a classic, full-employment, Walrasian CGE model. The different macro closure models range along the continuum.

Any CGE model in which factor markets are assumed to clear will generate full employment. Any macro model linked to such a CGE model can only have compositional effects—macro forces will affect the composition of final demand (C, I, and G), but not employment or GDP.

To extend the CGE model to incorporate aggregate employment effects requires some specification of neoclassical disequilibrium in the factor markets, which certainly stretches the Walrasian paradigm underlying the CGE model.

It is feasible to impose a variety of structuralist macro features on CGE models. While the justification for these features comes from macro theory outside the flow equilibrium structure of the CGE model, it is not necessary to incorporate features such as financial flows, asset markets, money, inflation, and interest rates directly into the CGE model. Many features of Keynesian demand-driven multiplier models can be accommodated within the flow-equilibrium structure of a CGE model.

The literature on macro closure indicates the robustness of an ecumenical modeling strategy, keeping macro models and CGE models separate, but linking them by imposing the results from the macro models on the CGE models.

The work program on integrating CGE and macro-financial models, while active, has proceeded very slowly, with fits and starts. Part of the problem is that the literature on macroeconomic theory and practice has not always been very helpful. Macro theorists came to recognize the deficiencies of demand-driven models with very weak supply sides, but then swung to the opposite extreme, building long-run, dynamic rational expectations models with strong roots in neoclassical growth theory, but with weak empirical foundations. The pendulum is again in motion, and there are now a number of examples of empirical dynamic macro models which incorporate simple CGE models to provide the supply side. From the CGE side, there is an active literature on dynamic CGE models which incorporate many features from the theoretical literature on dynamic models, including forward-looking expectations.[xxviii] Recent advances in modeling software have made the implementation of large, dynamic, empirical multisector models feasible. It seems to be a propitious time to link the two strands of work.

## NOTES

[i] In "type 2" input-output multiplier models, the factor accounts and the household account are left endogenous.

[ii] See Defourny and Thorbecke (1984) on path decomposition of SAM multipliers. See also Pyatt and Roe (1977), Chapter 4, and Robinson and Roland-Holst (1988), who decompose total multipliers into direct and indirect components.

[iii] Adding a labor supply function does not change the essential properties of this model. It is still a full-employment model, solving for a market wage that clears the labor market.

[iv] Armington (1969).

[v] See, for example, Dervis, de Melo, and Robinson (1982); de Melo and Robinson (1989); Devarajan, Lewis, and Robinson (1990, 1993); de Melo and Tarr (1992); and Thierfelder and Robinson (2002).

[vi] The theoretical properties of the real exchange rate in this model are worked out in Devarajan, Lewis, and Robinson (1993).

[vii] Expanding the "world" by adding more countries and solving for world prices endogenously in a multi-country CGE model does not make the interpretation of trade balances any easier. Such global models typically do not include explicit treatment of asset markets, and so specify and solve for only flow equilibria in world commodity markets. Most world trade models typically specify fixed national trade balances, recognizing that that they largely depend on the operation of asset markets that are outside the CGE structure.

[viii] This is an example of a macro "closure" of the CGE model. Other examples will be discussed below.

[ix] It is feasible, and sometimes done, to specify utility functions such as the extended linear expenditure system (ELES) that explicitly include savings, but there is still no tracking of the uses of the savings.

[x] See, for example, Lewis (1985) who introduced loanable funds markets into a CGE model to examine issues of financial repression and credit constraints.

[xi] There are exceptions in the public finance literature where government is treated as analogous to a household, with its own utility function. See Shoven, and Whalley (1992).

[xii] Examples of work expressing this view include Whalley and Yeung (1984), Bell and Srinivasan (1984), and Srinivasan (1982).

[xiii] Examples of work in this tradition include McKibbin and Sachs (1991); McKibbin and Wilcoxen (1998); Agénor and Montiel (1996); Borguignon, de Melo, and Suwa (1991); Thorbecke (1991). Robinson (1991) surveys models which incorporate asset markets in a dynamic CGE framework.

[xiv] Robinson and Tyson (1984) formalized the approach, and Powell (1981) describes applications using the Orani model of Australia.

[xv] See Bourguignon, de Melo, and Suwa (1991), Thorbecke (1991), and Robinson (1991) for a discussion of FSAMs and their use in integrated CGE/macro models. Taylor (2004), Chapter 2, uses an elegant integrated flow/asset SAM to provide an analytic framework for a discussion of structuralist macro models.

[xvi] For other discussions of the links, see Thorbecke (1985) and Robinson and Roland-Holst (1988).

[xvii] See, for example, Thorbecke (1998) and Hoffmann, Robinson, and Subramanian (1996).

[xviii] See, for example, Sen (1963), Taylor (1990), Rattso (1982), Robinson (1989, 1991), and Dewatripont and Michel (1987).

[xix] Malinvaud (1977) discusses the different notions of "equilibrium" in macro and general equilibrium models.

[xx] See Taylor (1983, 1990).

[xxi] This closure is named after Leif Johansen, who used it in the first CGE model (originally published in 1960). See Johansen (1974).

[xxii] A standard practice in structuralist macro models is to assume that the output price is set by some markup rule, rather than through profit maximization, and then output is demand driven. Again, see Taylor (1983, 1990).
[xxiii] See also de Melo and Robinson (1989).
[xxiv] Examples of applied models using this macro closure include Ahluwalia and Lysy (1981) and Devarajan and de Melo (1987). Devarajan and de Melo specify the exchange rate as numeraire and assume that the real exchange rate adjusts through changes in the aggregate price level.
[xxv] This is not quite true. Foreign savings are defined as fixed in "foreign" currency units. Changes in the exchange rate will change their value in domestic currency units.
[xxvi] See Lysy and Taylor (1980).
[xxvii] Adelman and Robinson (1988) compare the impact of this closure to the standard Keynesian closure in models of Brazil and Korea.
[xxviii] See, for example, Diao, Roe, and Yeldan (1999).

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